

RiskGuard AI

Road Accident Risk Assessment System

User Manual

Version 1.0 | June 2026

Author: Mubina Rustamova

mubinarustamova86@gmail.com

Table of Contents

- 1. Introduction
- 2. System Requirements
- 3. Installation & Setup
- 4. Logging In
- 5. Dashboard Overview
- 6. Performing a Risk Assessment
- 7. Reading Your Results
- 8. Downloading a PDF Report
- 9. Risk Comparison Tool
- 10. AI Chatbot
- 11. Assessment History
- 12. Insurance Quote Calculator
- 13. Admin Features
- 14. Troubleshooting
- 15. Glossary

1. Introduction

RiskGuard AI is a web-based road accident risk assessment system designed for insurance companies and underwriters. It uses a machine learning model trained on 12,000 synthetic driver records to predict a driver's risk profile, calculate a premium multiplier, and provide personalised recommendations for risk reduction.

The system provides:

- A risk score on a 0–100 scale with four categories: Low, Medium, High, Very High
- A premium multiplier (1.0× to 2.30×) for pricing decisions
- Per-factor risk contribution breakdown with an interactive chart
- An improvement roadmap showing how each change reduces the risk score
- A downloadable PDF report for each assessment
- An AI-powered chatbot for driver guidance (requires API key)
- A side-by-side driver comparison tool
- An admin dashboard with historical trends and model metrics
- Bilingual interface: English and Uzbek

2. System Requirements

Server / Developer Machine:

- Operating System: Windows 10/11, macOS 12+, or Ubuntu 20.04+
- Python 3.10 or later
- 4 GB RAM minimum (8 GB recommended for model training)
- 500 MB free disk space
- Internet connection (only required for AI chatbot feature)

End-User Browser (no installation required for users):

- Google Chrome 110+, Mozilla Firefox 110+, Microsoft Edge 110+, Safari 16+
- JavaScript must be enabled
- Minimum screen resolution: 1024 × 768

3. Installation & Setup

Step 1 — Create the .env file

In the `Source_Code/backend/` folder, create a plain text file named `.env` (note the leading dot). Add the following lines, replacing the placeholder values:

```
SECRET_KEY=any-random-secret-string-here ADMIN_USERNAME=admin  
ADMIN_PASSWORD=YourSecurePassword123 ANTHROPIC_API_KEY=sk-ant-... (optional)
```

```
EMAIL_USER= (optional Gmail address) EMAIL_PASS= (optional Gmail app password)
```

Note: Never share your .env file or commit it to version control.

Step 2 — Install Python dependencies

Open a terminal (Command Prompt or PowerShell on Windows) in the Source_Code/backend/ directory and run:

```
pip install -r requirements.txt pip install reportlab
```

Step 3 — Generate dataset and train the model (first time only)

This step generates 12,000 synthetic training records and trains the Gradient Boosting risk model. It takes approximately 30–60 seconds.

```
python setup.py
```

When complete, you will see a confirmation message and two new files appear in the model/ folder: **risk_model.pkl** and **model_metrics.pkl**.

Step 4 — Start the application

Option A — Windows one-click launcher (recommended):

```
Double-click run.bat
```

Option B — Manual start:

```
python app.py
```

The application will automatically find a free port starting from 5000 and display the URL in the terminal, e.g.: **http://127.0.0.1:5000**

4. Logging In

Open your browser and navigate to **http://127.0.0.1:5000**. You will be redirected to the login page.

Enter the **ADMIN_USERNAME** and **ADMIN_PASSWORD** you configured in the .env file, then click **Sign In**.

Security note: The login is rate-limited to 10 attempts per minute. After repeated failed attempts, the system will temporarily block the IP address. Sessions expire after 8 hours of inactivity.

5. Dashboard Overview

After logging in, you arrive at the **Dashboard**. It displays:

Summary Cards (top row):

- Total Assessments — cumulative count of all assessments performed
- Average Risk Score — mean score across all assessments (0–100)
- High Risk Count — number of assessments in the High or Very High category
- Average Premium Multiplier — mean multiplier across all assessments

Charts:

- Risk Category Distribution — doughnut chart showing Low/Medium/High/Very High split
- Score Over Time — line chart of the last 30 assessments
- Top Risk Factors — bar chart of most common contributing factors

Model Performance Panel:

- Shows the current model type (Gradient Boosting Tuned)
- Accuracy, F1-Score, Precision, and Recall metrics
- Training / test set sizes

6. Performing a Risk Assessment

Click **Risk Assessment** in the navigation menu. You will see a form with 12 input fields organised into three sections.

Section A — Driver Profile

Fill in the following fields:

- **Driver Age:** Enter the driver's age in years (18–80)
- **Driving Experience:** Years of active driving experience (0–60)
- **Gender:** Select Male or Female

- **Marital Status:** Select Single, Married, or Divorced
- **Credit Score:** Enter the driver's credit score (300–850). A score of 700+ removes the credit surcharge.

Section B — Vehicle Details

- **Vehicle Type:** Choose from Sedan, SUV, Sports, Truck, Van, Motorcycle. Motorcycle carries the highest risk penalty (+18 pts); Sedan is the baseline.
- **Vehicle Age:** Age of the vehicle in years (0–25)
- **Primary Location:** Where the driver most commonly drives — Urban (+7 pts), Highway (+4 pts), Rural (+2 pts), Suburban (baseline)

Section C — Driving Behaviour

- **Annual Mileage:** Estimated miles driven per year (3,000–55,000)
- **Night Driving %:** Percentage of driving done at night (0–100)
- **Previous Accidents (3 yrs):** Number of at-fault accidents in the last 3 years (0–4). This is the strongest risk predictor (+14 pts per accident).
- **Traffic Violations:** Number of violations in the past 12 months (0–5). Each violation adds +5 pts to the risk score.

Once all fields are completed, click **Calculate Risk** to run the assessment.

7. Reading Your Results

The results page displays the following sections:

Risk Score Gauge

A large circular gauge showing your score (0–100). The colour indicates category:

- **Green (0–25):** Low Risk — excellent profile
- **Amber (26–50):** Medium Risk — acceptable with some concerns
- **Red (51–75):** High Risk — significant risk factors present
- **Purple (76–100):** Very High Risk — immediate action recommended

Premium Multiplier

The multiplier applied to the base insurance premium:

- Low Risk: 1.00× (no surcharge)
- Medium Risk: 1.35× (+35%)
- High Risk: 1.75× (+75%)
- Very High Risk: 2.30× (+130%)

Claim Probability

The estimated probability (0–100%) that this driver will make a claim. Derived from the ML model's class probabilities for High and Very High risk categories.

Factor Contributions Chart

A horizontal bar chart showing which factors are contributing risk (red bars) or providing protection (green bars). Each bar shows the magnitude of the factor's contribution to the final risk score.

Improvement Roadmap

A prioritised list of up to 5 actions the driver can take to reduce their score. Each action shows the estimated score saving based on counterfactual simulation through the ML model.

Personalised Recommendations

Specific advice tailored to the driver's risk factors — e.g., telematics device installation, defensive driving courses, or credit score improvement tips.

8. Downloading a PDF Report

From the results page, click the **Download PDF Report** button. A styled PDF will be generated in-memory and downloaded to your browser.

The PDF includes: risk score gauge, factor breakdown table, improvement roadmap, recommendations, and a summary of the driver profile used for the assessment.

Each assessment is also saved to the database and can be retrieved later from **Assessment History** in the navigation menu.

9. Risk Comparison Tool

Navigate to **Tools** → **Compare Drivers**. Enter two driver profiles and click **Compare**. The page displays:

- Side-by-side risk scores and categories for both drivers
- A head-to-head bar chart of key metrics
- A factor-by-factor comparison table with a winner indicator
- Class probability charts for each driver

Use cases: Compare a named driver vs main policyholder; evaluate a young driver vs experienced driver; compare two vehicle types for the same driver.

10. AI Chatbot

Navigate to **Tools** → **AI Assistant**. The chatbot (RiskBot) can answer questions about risk factors, premiums, coverage, safe driving, and more.

AI Mode: If an ANTHROPIC_API_KEY is configured in .env, the chatbot uses Claude Haiku for natural language understanding. It maintains context across the last 10 conversation turns.

Fallback Mode: Without an API key, the chatbot uses regex-based intent matching across 20+ intent categories. It handles questions about premiums, telematics, NCB, young drivers, credit scores, and more.

The chatbot responds in the same language as the user's input (English or Uzbek).

11. Assessment History

Click **History** in the navigation menu to view all past assessments. You can filter by date range and risk category. Each row links to the full results page, where you can re-download the PDF report.

12. Insurance Quote Calculator

Navigate to **Tools** → **Get Quote**. Enter a base premium (e.g., £800/year). The tool applies the driver's risk multiplier and any No-Claims Bonus (NCB) discount to calculate the estimated annual and monthly premium.

NCB discounts available: 0 yrs (0%), 1 yr (20%), 2 yrs (35%), 3 yrs (45%), 4 yrs (52%), 5+ yrs (60%).

13. Admin Features

- **Dashboard:** Summary statistics and trend charts for all assessments
- **Model Report:** Full ML metrics — accuracy, F1, precision, recall, confusion matrix, feature importance, and comparison across all 4 models
- **Batch Upload:** Upload a CSV file of multiple driver profiles for batch risk scoring (downloads results as CSV)
- **Export CSV:** Download the full assessment history as a CSV file
- **API Documentation:** Built-in docs page listing all REST endpoints
- **Language Toggle:** Switch UI between English and Uzbek

14. Troubleshooting

App won't start

- Ensure `.env` file exists in the backend folder with all required keys
- Run `'python setup.py'` first if `risk_model.pkl` is missing
- Check that port 5000 is not already in use (try 5001, 5002)

'Model not found' error

- Run `'python setup.py'` to generate data and train the model

PDF download fails

- Install reportlab: `pip install reportlab`

Chatbot not responding intelligently

- Add a valid `ANTHROPIC_API_KEY` to `.env` to enable Claude AI mode
- Without the key, the chatbot uses pattern matching — it still works for common questions

Login blocked

- Rate limit is 10 attempts/minute. Wait 60 seconds and try again

Check the log file

All errors are logged to **app.log** in the backend folder. This is the first place to look when diagnosing issues.

15. Glossary

Risk Score: A number from 0 to 100 representing the driver's accident risk level. Higher = riskier.

Premium Multiplier: A factor applied to the base insurance premium based on risk category. Ranges from 1.0× (Low) to 2.30× (Very High).

Gradient Boosting: The machine learning algorithm selected as the production model. It builds an ensemble of decision trees sequentially, each correcting the errors of the previous.

GridSearchCV: Automated hyperparameter tuning technique that tests all combinations of specified parameters using 5-fold cross-validation.

Feature Importance: A measure of how much each input variable contributes to the model's predictions. Previous accidents ranks highest in this system.

Claim Probability: The estimated likelihood (0–100%) that a driver will make an insurance claim, derived from the ML model's class probabilities.

NCB (No-Claims Bonus): A discount applied to the premium for each consecutive claim-free year. Can reach up to 60% after 5+ years.

Telematics: Technology (black box device or app) that monitors actual driving behaviour in real time. Usage-Based Insurance (UBI) uses telematics data to set premiums.

CSRF: Cross-Site Request Forgery — a security attack. Flask-WTF CSRF protection is active on all forms to prevent this.

SQLite: A lightweight, file-based relational database. No separate server needed. The assessments.db file is created automatically on first run.

OneHotEncoding: Converts categorical text values (e.g., 'sedan', 'urban') into numeric columns that the ML model can process.

StandardScaler: Normalises numeric features to have mean=0 and std=1, ensuring no single feature dominates due to scale differences.